

FINAL COMPREHENSIVE LIVE DEMONSTRATION PLAN

Project ID: SIH26034
Product: NyayaDrishti-LM
Audience: Smart India Hackathon 2026 Evaluation Jury (DoCA & Domain Experts)
Status: FROZEN

1. The Three-Tier Demo Safety Architecture

To guarantee that the live evaluation never fails regardless of hardware glitches, lighting variations, or podium network constraints, NyayaDrishti-LM implements a **3-Tier Demo Architecture**:

3-TIER LIVE DEMO STRATEGY		
TIER 1: PRIMARY LIVE DEMO EMERGENCY STATIC GOLDEN (Interactive Physical Camera Merkle Dossiers)	TIER 2: BACKUP LOCAL STREAM (Virtual Video Stream)	TIER 3: (Pre-Computed
- Real physical FMCG packages loading of 5 golden - ArUco reference card placed records in SQLite - Live laptop webcam / phone signed PDFs - Real-time blur/glare HUD full UI, Merkle - Instant PDF report output rule citations	- High-resolution recorded video stream of real packs - Simulates live camera feed - Evaluates live ONNX models - Immune to podium glare	- Instant inspection - Pre-generated - Demonstrates hashing, and

2. The 3-Minute Live Judging Script

MINUTE 1: The Problem & The Scientific Litmus Test (0:00 - 1:00)

- **Speaker:** Team Leader
- **Action:** Hold up a real biscuit carton and an uncalibrated smartphone.
- **Pitch:**

"Respected Judges, India has 3,000 Legal Metrology officers regulating billions of packages across 12 million retail shops. Today, an officer spends 8 minutes per pack using a magnifying loupe and manual arithmetic. Less than 0.1% of packages are ever inspected.

Amateur teams attempt to point a camera and prompt an LLM to 'judge' the label. But as metrology experts know, a 2D camera sees pixels, not physical millimeters. Claiming to measure font height without a calibration scale violates projective geometry.

*We present **NyayaDrishti-LM**: an offline-capable, hybrid perception-verification system that solves the millimeter problem with mathematical rigor and enforces the Legal Metrology Rules, 2011 deterministically."*

MINUTE 2: Live Ingestion, Calibration & Violation Detection (1:00 - 2:00)

- **Speaker:** Member 1 & Member 6
- **Action:** Place a standard reference card next to the biscuit box. Capture the front panel on camera.
- **UI Demonstration:**
 1. *Optical Quality Gate:* Show the green **FRAME_OPTIMAL** indicator. Flash a light to show the instant red **SPECULAR_GLARE** warning and recovery guidance.
 2. *Metric Scale Rectification:* Show the ArUco marker corners detected in real time, perspective rectified, and scale factor $S = 0.052 \text{ mm/pixel}$ established.
 3. *Multilingual OCR:* Show DBNet++ polygons tightly wrapping text in English and Hindi.
 4. *Rule Engine Execution:*
 - *Net Quantity:* Reads **200 gms** → **FAIL:** Highlights banned unit 'gms' under Section 11 & Rule 12.
 - *Principal Display Panel Area:* Computes $A = 144 \text{ cm}^2$ (Row 3: requires $\geq 2.5 \text{ mm}$).
 - *Physical Font Measurement:* Measures glyph height at $1.84 \text{ mm} \pm 0.08 \text{ mm}$ → **FAIL:** Displays deficit of 0.66 mm citing Table-I.

- **Unit Sale Price:** MRP = ₹80, declared USP = ₹0.60/g → **FAIL:** Flags arithmetic mismatch ($200 \times 0.60 = ₹120 \neq ₹80$).

MINUTE 3: Human-in-the-Loop, BSA Evidence & PDF Dossier (2:00 - 3:00)

- **Speaker:** Member 4 & Member 5
- **Action:** Navigate to the Side-by-Side Review HUD and click *Generate Statutory Dossier*.
- **Pitch:**

"Notice that our AI did not issue a fine autonomously. Under the Legal Metrology Act and the Jan Vishwas Act, 2023, statutory power rests exclusively with the human officer. The inspector reviews the side-by-side rectified crop, validates the findings, and clicks sign-off.

Instantly, our system builds an immutable SHA-256 Merkle DAG conforming to Section 63 of the Bharatiya Sakshya Adhiniyam, 2023, and compiles a court-ready, tamper-evident Inspection Dossier citing exact Gazette GSR notifications and recommending a statutory Improvement Notice.

All of this runs 100% offline on standard laptop CPUs in under 1.5 seconds with zero cloud API costs. This is not a research toy; it is an operational enforcement tool for the Department of Consumer Affairs."

3. Golden Test Cases for Demonstration

SKU ID	Product Category	Intended Demonstration Point	Expected System Output
SKU-DEMO-01	Premium Biscuit Carton	Font height violation (1.84 mm vs 2.50 mm req) + Banned unit gms	VIOLATION_FLAG on Table-I Row 3 and Section 11.
SKU-DEMO-02	Ready-to-Eat Curry Pouch	USP mathematical mismatch + Incomplete Consumer Care (missing email)	VIOLATION_FLAG on Rule 6(1)(f) and Rule 6(1)(g).
SKU-DEMO-	Packaged Bottled	Fully compliant modern packaging (GSR 779	VERIFIED_COMPLIANT (PASS) across all statutory clauses.

SKU ID	Product Category	Intended Demonstration Point	Expected System Output
03	Water	USP compliant, valid font)	
SKU-DEMO-04	Borderline FMCG Soap Box	Font height within measurement uncertainty (2.46 mm $\pm$ 0.08 mm)	REQUIRES_HUMAN_REVIEW with prompt for caliper verification.
SKU-DEMO-05	Reflective Chips Pouch	Harsh specular glare bloom over Net Quantity statement	UNABLE_TO_VERIFY (SPECULAR_GLARE) with camera tilt prompt.